Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

Task 2's modelling framework was executed as a baseline check (500 customers, 16.0% delinquent): logistic regression, gradient boosting and a dummy baseline all scored at the no-skill floor (PR-AUC 0.16-0.15 vs a 0.16 base rate), confirming Task 1 - no recorded variable is yet a validated predictor. The segments below are the strongest observed (descriptive) gradients: fit for a controlled pilot, not automated decisions. Per the Dataset Description Guide, Missed_Payments covers 12 months and Month_1-6 cover 6 months; neither related to the target (p = 0.56).

- Heavy debt burden: Debt_to_Income_Ratio above 40% delinquent at 18.9% (14/74) - nearly double the 9.8% (8/82) below 20% (strongest gradient; not significant, p = 0.10).

- Income instability: unemployed delinquent at 19.4% (18/93) vs 11.5% (10/87) retired; Employment_Status needed label clean-up ('EMP'/'employed', 245 rows).

- Product concentration: Business cardholders delinquent at 21.3% (23/108) - highest of five Credit_Card_Type tiers - vs 11.8% (9/76) Platinum.

Top 3 Risk Factors (for easy communication):

- 1. Debt-to-income ratio above 40% - roughly double the rate of the lowest band.
- 2. Unemployed customers - the highest delinquency of any employment group (19.4%).
- 3. Business card tier - the highest of all product tiers (21.3%).

All three are descriptive candidates; none is statistically validated (Task 2 go/no-go gate applies).

Key Insights Summary Table:

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Debt burden roughly doubles delinquency	Accounts with DTI > 40% (n = 74, ~15% of book)	Debt_to_Income_Ratio; Loan_Balance	Priority cohort for proactive restructuring outreach
Income instability elevates risk	Unemployed customers (n = 93)	Employment_Status; Income	Route to hardship and collections-support programmes
Risk concentrated in one product tier	Business cardholders (n = 108)	Credit_Card_Type	Product-level review of limits and underwriting

2. Recommendation Framework

The recommendation converts the strongest observed EDA insight into a SMART intervention, respecting the Task 2 finding that no model may yet be deployed.

- **Restated Insight:** DTI above 40% delinquent at 18.9% vs 9.8% below 20% - the strongest observed gradient, not yet statistically confirmed.
- **Proposed Recommendation:** Run a controlled 90-day Collections pilot: proactive, supportive outreach (hardship restructuring, limit review) to every account with DTI above 40%, against a matched control.
    - **Specific:** Target the DTI > 40% cohort (74 accounts, ~15% of book); matched control from the 20-30% band.
    - **Measurable:** >= 15% relative delinquency reduction vs control at day 90; track contact, cure and complaint rates and cost per account, monthly.
    - **Actionable:** Existing Collections team and CRM; rule-based campaign, no model deployment (Task 2 gate).
    - **Relevant:** Targets the highest-observed-delinquency segment, supporting credit-loss reduction while data remediation proceeds.
    - **Time-bound:** Launch within 4 weeks, run 90 days, scale-or-stop decision at day 91.
- **Justification and Business Rationale:** Lowest-cost way to test the top observed gradient under controlled conditions: if confirmed, Geldium gains a validated, explainable targeting rule; if not, the gate redirects effort to data remediation instead of acting on noise. Transparent, feasible with current resources, and avoids model investment while predictors remain unvalidated (PR-AUC at the 0.16 no-skill floor).

3. Ethical and Responsible AI Considerations

With no validated signal in the data, the main ethical risk is acting on noise as if it were fact. Two fairness risks and mitigations:

- **Fairness risk 1 - proxy discrimination:** Location and Employment_Status correlate with ethnicity and socioeconomic status; punitive use (limit cuts, rejections) could disadvantage protected groups. Mitigation: DTI targeting for supportive offers only, exclude Location from scoring, audit pilot outcomes by age, employment group and region.
- **Fairness risk 2 - vulnerable-group impact:** Unemployed customers already show the highest delinquency (19.4%); aggressive collections could deepen hardship. Mitigation: human-in-the-loop review, hardship-programme routing, plain-language reason codes (e.g., 'your debt obligations are close to your income').
- **Explainability for non-technical stakeholders:** The recommendation is a transparent DTI-band rule, not a black box; any future model must give per-decision reason codes. Predictions are communicated as probabilities with uncertainty - the executed baseline showed no better than chance, so no automated adverse decisions are justified.
- **Responsible and transparent AI use:** Transparency (pilot limits documented, including 7.8% missing Income affecting banding), accountability (named owner, model card, stop rule), data privacy (income/debt data minimised), and a fairness re-audit before scale-up.